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**Information Assurance and Security Enhancement
Plan for HG Esports Cafe**

**A Final Term Requirement for The Course
INFORMATION AND ASSURANCE 1**

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TABLE OF CONTENTS

Project Name	5
Project Description	5
Project Proponents	5
Organizational Background	5
Current Information Security Status	6
Information Security Assessment	7
Project Schedule	7
Project Costs	8
1. Inventory of Authorized and Unauthorized Devices	10
2. Inventory of Authorized and Unauthorized Software	10
3. Secure Configurations for Hardware and Software	11
4. Malware Defenses	13
5. Application Software Security	13
6. Wireless Access Control	14
7. Data Recovery Capability	15
8. Information Security Personnel Requirements and Their Qualifications	16
9. Secure Configurations for Network Devices such as Firewalls, Routers, and Switches	17
10. Controlled Use of Administrative Privileges	18
11. Boundary Defense	20
12. Maintenance, Monitoring, and Analysis of Audit Logs	23
13. Controlled Access Based on the Need to Know	25



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14. Data Protection	26
15. Incident Response and Management	27
Appendix A. Vulnerability Assessment Report	30
Appendix B. Interview Transcript	45
Appendix C. Photo Documentaton.....	47
Appendix D. Audit Report.....	49



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LIST OF TABLES

Table 1. Project Implementation Timeline	7
Table 2. Implementation Cost Estimate	8
Table 4. Current Status and Proposed Procedures	12
Table 5. Proposed Technologies and Estimated Cost	13
Table 6. Current Status and Proposed Procedures	13
Table 7. Proposed Technologies and Estimated Cost	14
Table 8. Current Status and Proposed Procedures	14
Table 9. Proposed Technologies and Estimated Cost	15
Table 10. Current Status and Proposed Procedures	15
Table 11. Proposed Technologies and Estimated Cost	16
Table 12. Proposed Functional Roles	16
Table 13. Proposed Training Program	17
Table 14. Proposed Role-Based Access Control	19
Table 15. Current Status and Proposed Procedures	24
Table 16. Proposed Technologies and Estimated Cost	24
Table 17. Current Status and Proposed Procedures	25
Table 18. Proposed Technologies and Estimated Cost	26
Table 19. Current Status and Proposed Procedures	26
Table 20. Proposed Technologies and Estimated Cost	27
Table 21. Current Status and Proposed Procedures	28
Table 22. Incident Response Framework	28
Table 23. Proposed Technologies and Estimated Cost	29



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Project Name

Information Assurance and Security Enhancement Plan for HG Esports Cafe

Project Description

This project is proposed to improve the current information security condition of HG Esports Cafe by addressing the vulnerabilities identified during the assessment. The organization currently applies some security controls, but these are only partially implemented and not formally documented.

The plan focuses on improving how devices, systems, and networks are managed and protected. It also introduces clear procedures to reduce risks such as malware infection, system downtime, and unauthorized access. Both technical solutions and proper security practices will be applied to support the organization's daily operations.

Project Proponents

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Organizational Background

Company Name

HG Esports Cafe

Number of Years in Operation

The organization is newly established and still developing its systems and operations.

Number of Employees

The organization has four (4) employees managing daily operations.

Nature of Business

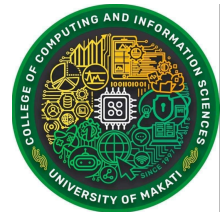
HG Esports Cafe provides computer rental services, online gaming access, and internet-related services to customers.



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Business Processes requiring Information Assurance and Security

- Customer computer usage and session management
- Network and internet service delivery
- Billing and transaction processing
- Server-based operations
- Handling of user session data and business records

These processes require proper security to ensure that systems remain reliable and that data is protected from unauthorized access or loss.

Current Information Security Status

Based on the assessment, HG Esports Cafe is currently at a developing level of information security. Some controls are present, but they are not fully implemented or formally managed.

Strengths

- Systems and software are regularly updated
- Employees have assigned access levels
- Reboot-to-restore system is used after each session
- Separate network is provided for Wi-Fi users
- Physical equipment is secured

These practices help reduce basic risks and support daily operations.

Weaknesses

- No firewall is implemented
- No UPS system for power backup
- USB devices are not controlled
- Users are allowed to download files
- Server is exposed within the area
- No formal security policies are documented
- Network depends on a single router
- Limited IT personnel

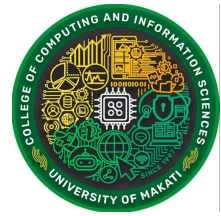
These weaknesses increase the risk of security incidents and system failures.



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Information Security Assessment

Threats

Possible threats that may affect the organization include:

- Malware infection through USB devices
- Unauthorized downloads by users
- Network failure due to single router setup
- Power interruption causing data loss
- Physical access to the exposed server
- System downtime due to hardware issues

Vulnerabilities

The following weaknesses were identified:

- No firewall or advanced network protection
- Unrestricted use of USB devices
- Lack of UPS system
- Absence of formal policies
- Use of non-enterprise server hardware
- Limited IT staff
- Uncontrolled downloading of files

Risk Assessment

- High Risk: Malware infection, system downtime, USB-related threats, server exposure
- Critical Risk: Server reliability issues and lack of network redundancy
- Low Risk: Updated systems and use of multiple ISPs

These risks show that immediate improvements are needed in both technical and operational areas.

Project Schedule

Table 1. Project Implementation Timeline



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Phase	Activity	Duration
Phase 1	Planning and Policy Development	2 weeks
Phase 2	Procurement of Security Tools	2 weeks
Phase 3	Implementation and Configuration	3 weeks
Phase 4	Testing and Monitoring	2 weeks
Phase 5	Training and Documentation	1 week

Total Duration: **10 weeks**

The schedule allows the organization to gradually apply improvements while ensuring proper testing and staff preparation.

Project Costs (Estimated)

Table 2. Implementation Cost Estimate

Item	Purpose	Cost
Firewall Device (e.g., TP-Link ER605 upgrade or dedicated firewall)	To protect the network from external threats and unauthorized access	₱30,000 – ₱60,000
UPS Units (10–15 units for server + critical PCs)	To prevent system shutdown and data loss during power interruption	₱30,000 – ₱75,000
Endpoint Security Software (70 PCs) (e.g., ESET Endpoint Security)	To detect and prevent malware, ransomware, and unauthorized access	₱140,000 – ₱350,000 / year



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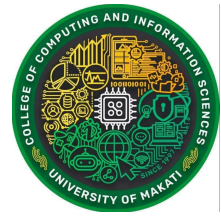
Device Inventory Tools (e.g., ManageEngine AssetExplorer)	To track and manage hardware, software, and network devices	₱15,000 – ₱50,000 / year
Training Materials / Awareness Program	To improve staff awareness on cybersecurity risks and safe practices	₱5,000 – ₱15,000

Table 3. Overall Project Cost Summary

Category	Cost
Initial Setup Cost	₱80,000 – ₱150,000
Annual Cost (Software & Tools)	₱155,000 – ₱400,000 / year
Total:	₱235,000 – ₱550,000 / year

The proposed cost allocation is based on the actual infrastructure of HG Esports iCafe, which operates approximately seventy (70) computer units supported by a single server and multiple internet connections. The absence of key security controls such as firewall protection, UPS systems, and clearly defined endpoint security increases the organization's exposure to cyber threats and operational disruptions.

Investing in firewall devices, endpoint protection, and power backup systems will significantly reduce risks such as malware infection, unauthorized access, and data loss due to sudden outages. Additionally, asset management tools and staff training will improve monitoring capabilities and strengthen overall security awareness, ensuring a more resilient and secure operational environment.



Proposed Information Assurance and Security Plan

1. Inventory of Authorized and Unauthorized Devices

Current Status

The organization has multiple devices, including 70 computers, a server, and network equipment. However, there is no formal system to track or manage these devices. USB devices are also freely used without monitoring.

Proposed Procedures

- Maintain a complete list of all hardware devices
- Assign identification labels to each device
- Monitor and restrict the use of USB devices
- Perform regular inventory checking

These steps ensure that all devices are accounted for and properly controlled.

Proposed Technologies

- Simple asset inventory system (software or spreadsheet)
- USB control tools

2. Inventory of Authorized and Unauthorized Software

Current Status

Although systems are updated, users are allowed to download files, and there is no system to monitor installed software.

Proposed Procedures

- Define and allow only approved software
- Restrict unauthorized downloads
- Perform regular software checking
- Monitor user activity

These procedures reduce the risk of installing harmful software.



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Proposed Technologies

- Antivirus or endpoint protection software
- Application control tools

3. Secure Configurations for Hardware and Software

Current Status

The systems are updated, but there is no firewall, no UPS, and the server is not fully secured. The network also relies on a single router.

Proposed Procedures

- Install firewall protection
- Secure the server in a restricted location
- Regularly review system configurations
- Create formal security policies
- Apply stricter user access controls

These procedures help improve overall system protection and reduce risks.

Proposed Technologies

- Firewall device
- UPS units
- Basic intrusion detection tools
- Standard system configuration settings

Additional Recommended Improvements

- Develop formal security policies
- Create basic incident response procedures
- Conduct staff training
- Improve network reliability
- Upgrade server hardware

These improvements support long-term security and better system management.



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4. Malware Defenses

Control the installation, spread, and execution of malicious code.

Implements layered malware defense using antivirus, USB restrictions, safe download controls, and DNS filtering, with system restore as backup protection to enhance security while keeping costs low.

Table 4. Current Status and Proposed Procedures

Area	Current Status	Proposed Procedures
Endpoint Protection	No centralized antivirus; relies on reboot-to-restore	Deploy antivirus on all PCs; enable real-time protection; daily quick scans; weekly full scans; assign staff to monitor alerts
USB Control	Unrestricted USB usage	Disable autorun; restrict USB access to staff control; allow temporary use of customer flash drives only with staff approval and supervision (e.g., staff enables access for a limited time or copies files on behalf of the user); monitor all USB activity.
Unsafe Downloads	Users can download any files	Allow downloads but enforce controls; use DNS/web filtering to block known malicious sites; restrict or warn on high-risk file types (e.g., .exe, .bat); display security prompts/reminders before downloads; automatically scan all downloaded files with antivirus; limit downloads to designated folders that are cleared after each session; monitor activity and flag suspicious behavior.
System Protection Layer	Uses reboot-to-restore only	Retain as secondary defense; reapply clean system image monthly



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Table 5. Proposed Technologies and Estimated Cost

Area	Proposed Technologies	Estimated Cost
Endpoint Protection	Microsoft Defender Antivirus/Antivirus	Free
USB Control	Local Group Policy Editor (USB restriction policies)	Free
Web/DNS Filtering	OpenDNS or Cloudflare DNS (malware blocking)	Free
System Restore	Deep Freeze (centralized version recommended)	₱2,000–₱3,000 per PC

5. Application Software Security

Manage the security life cycle of all software.

Enhances application security through controlled software installation, structured patch management, standardized system images, and automated updates to ensure consistency, reduce vulnerabilities, and maintain secure, up-to-date systems.

Table 6. Current Status and Proposed Procedures

Area	Current Status	Proposed Procedures
Software Control	No strict control over installed apps	Create approved software list; remove unauthorized apps; restrict installations to admin only
Patch Management High	Updates done but not structured	Schedule weekly updates; test on selected PCs before full rollout; maintain update log
System Standardization	PCs may have inconsistent setups	Create a master system image; reapply monthly or when needed
Game/App Updates	Manual and inconsistent	Enable auto-updates; assign staff schedule for monitoring updates



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Table 7. Proposed Technologies and Estimated Cost

Area	Proposed Technologies	Estimated Cost
Software Restriction	Local Group Policy Editor (block installers, enforce admin rights)	Free
Patch Management	Windows Update for Business	Free
Disk Imaging	Macrium Reflect Free or Clonezilla	Free
Game Platforms	Steam / Riot Client auto-update features	Free

6. Wireless Access Control

Manage the processes and tools used to track, control, prevent, and correct the security use of wireless local area networks.

Improves wireless security through VLAN segmentation, stronger Wi-Fi encryption, device monitoring, and business-grade access points to ensure controlled access, better performance, and protection against unauthorized users.

Table 8. Current Status and Proposed Procedures

Area	Current Status	Proposed Procedures
Network Segmentation	Flat network	Separate admin and customer networks using VLANs to isolate traffic and improve security
Wi-Fi Security	Basic password setup	Use strong WPA2/WPA3 and change them regularly
Device Monitoring	No monitoring of connected users	Regularly check connected devices and block any unknown devices
Wi-Fi Infrastructure	Basic setup	Use business-grade access points with bandwidth control and client isolation



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Table 9. Proposed Technologies and Estimated Cost

Area	Proposed Technologies	Estimated Cost
Router & VLAN	TP-Link ER605 (configure VLANs, access control)	Free
Network Monitoring	TP-Link Omada Controller	Free
Access Points	TP-Link Omada EAP225 (2–4 units recommended)	₱12,000–₱24,000
Wi-Fi Security	Built-in WPA2/WPA3, guest network, bandwidth control	Free

7. Data Recovery Capability

The processes and tools used to properly back up critical information with a proven methodology for timely recovery of it.

Table 10. Current Status and Proposed Procedures

Area	Current Status	Proposed Procedures
Backup System	No formal backup	Implement regular daily and weekly backups, with assigned staff in charge
Backup Storage	No storage solution	Store backups in external hard drives kept in a secure location
Offsite Backup	None	Sync critical data to cloud storage weekly
Power Protection	No UPS	Use a UPS for the server and backup devices to prevent data loss during power outages



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Table 11. Proposed Technologies and Estimated Cost

Area	Proposed Technologies	Estimated Cost
Backup Software	Windows Backup and Restore	Free
External Storage	Seagate (2–4TB HDD)	₱5,500–₱10,000
Cloud Backup	Google Drive or Microsoft OneDrive	₱100–₱300/month
Power Backup	APC / CyberPower UPS (650–1200VA)	₱3,000–₱7,000

8. Information Security Personnel Requirements and Their Qualifications

Security Skills Assessment and Appropriate Training to Fill Gaps

The effective defense of HG Esports Café's enterprise environment necessitates the establishment of defined security roles supported by appropriate qualifications and continuous training.

Table 12. Proposed Functional Roles

Role	Responsibilities	Required Qualifications
IT Administrator	Manage PCs, network, router, server, user accounts, updates	Degree or training in Information Technology, knowledge of networking and systems administration (Windows Server hardening, Active Directory security, and backup/recovery protocols).
Security Officer	Monitor threats, enforce policies, incident response, audits	Background in cybersecurity, familiarity with access control and threat mitigation.
Technical Support Staff	Assist customers, monitor downloads, inspect devices	Technical troubleshooting skills, cybersecurity awareness.
Operations Manager	Oversight of compliance, resource allocation, business	Managerial experience and understanding of operational



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	continuity planning	risk.
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Security Skills Gap Assessment

Based on the vulnerability assessment findings, the organization exhibits capability gaps in areas such as firewall deployment and configuration management, malware prevention and removable media control, incident response planning and documentation, privileged access management, network segmentation and redundancy planning, and security awareness and policy enforcement.

Moreover, a skills assessment should be conducted using the NIST NICE Cybersecurity Workforce Framework in assigning IT Admin, Security Officer, and Support Staff competencies.

Table 13. Proposed Training Program

Training Topic	Intended Participants	Frequency
Cybersecurity Awareness and Acceptable Use	All employees	Quarterly
Firewall and Router Administration	IT Administrator	Semi-annual
Malware Detection and Incident Handling	Technical staff	Quarterly
Data Privacy and Confidentiality	All employees	Annual
Incident Escalation and Reporting Procedures	All employees	Quarterly

9. Secure Configurations for Network Devices such as Firewalls, Routers, and Switches

Secure configuration management is fundamental in protecting the network infrastructure of HG Esports Cafe. While the organization utilizes multiple internet service providers and managed network devices, these assets must be governed by formal security standards.



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Firewall Security Controls

The organization should deploy a dedicated firewall solution to regulate inbound and outbound traffic. The firewall should be configured to support Intrusion Prevention System (IPS) capabilities, web content filtering, malware traffic detection, Denial-of-Service (DoS) mitigation, and geographic blocking of suspicious traffic sources.

Router Security Standards

All routers should be hardened through measures, including, replacement of default administrative credentials, use of strong password policies, disabling remote management when unnecessary, restricting administrative access to authorized devices only, and regular firmware and security updates

Switch Security Controls

Managed switches should be configured to enforce:

- Deactivation of unused ports
- Port security to prevent unauthorized device connections
- Segmentation through Virtual Local Area Networks (VLANs) for: Gaming workstations, Administrative/server systems, Guest wireless access

Configuration Backup and Logging

All device configurations should be backed up monthly in encrypted storage. In addition, system logs should be enabled to record failed login attempts, administrative configuration changes, unusual bandwidth consumption, unauthorized connection attempts.

10. Controlled Use of Administrative Privileges

Administrative privileges present a significant security risk when improperly assigned or unmanaged. The HG Esports Cafe operates with seventy endpoints, hence, strict controls over privileged accounts are necessary to prevent misuse and unauthorized changes.

Principle of Least Privilege



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Administrative access should be granted strictly according to job responsibilities.

Table 14. Proposed Role-Based Access Control

Role	Access Right	Restrictions	Rationale
Operations Manager	Access to business reports, billing summaries, personnel schedules, audit logs, and policy approvals.	No direct access to firewall configuration, server administration, or endpoint system files unless authorized.	Ensures management oversight while minimizing unnecessary technical privileges.
IT Administrator	Full access to network devices, server systems, user account management, software deployment, backups, and security configurations.	Access limited to authorized maintenance tasks; activities must be logged and reviewed.	Required to maintain infrastructure, implement updates, and respond to technical incidents.
Information Security Officer	Access to security logs, audit trails, firewall alerts, antivirus console, incident reports, and compliance documentation.	No authority to modify billing data or customer transactions without approval.	Supports independent monitoring, risk management, and enforcement of security controls.
Technical Support Staff	Access to workstation diagnostics, timer system support tools, approved software troubleshooting utilities, and peripheral device management.	No access to router settings, privileged accounts, financial records, or security policy settings.	Enables efficient technical assistance while preventing unauthorized system changes.
Front Desk Personnel	Access to billing system, session timer controls, customer	No access to server files, network settings, security	Restricts personnel to customer service functions and



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	account registration, and payment processing modules.	logs, administrative utilities.	or reduces internal misuse risk.
Customer User	Access only to assigned gaming workstation, approved applications, internet browsing, and temporary user session storage.	No administrative privileges, no access to system settings, server resources, network devices, or other user data.	Ensures customers can use services while protecting systems and preserving confidentiality.

Account Governance Measures

The organization should implement the following controls:

- Individual administrator accounts for each authorized employee
- Prohibition of shared administrative credentials
- Strong password enforcement
- Multi-factor authentication for privileged accounts
- Immediate revocation of access upon employee separation

Monitoring and Review

All privileged activities should be logged quarterly, including:

- Software installation
- User account creation or deletion
- Router and firewall access
- Policy modifications

11. Boundary Defense

Boundary defense refers to the protection of the organization's internal network against unauthorized access, malicious traffic, and threats originating from external or lower-trust networks. For HG Esports Cafe, this control is essential because gaming workstations, administrative systems, customer devices, and internet service provider connections operate within the same business environment. To ensure effective



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protection, the organization should implement layered boundary security controls supported by appropriate tools and technologies.

A. Network Segmentation Strategy

The internal network should be logically separated into distinct trust zones to minimize lateral movement of threats and unauthorized access.

Proposed Network Zones:

1. Administrative and Server Network
2. Gaming Workstation Network
3. Guest Wi-Fi Network
4. CCTV and Auxiliary Device Network

Proposed Tools / Technologies:

- Managed Switches with VLAN Support and Router with VLAN Routing Capability
- Network Access Control (NAC) – Cisco ISE (enterprise option) or simplified MAC filtering for SMEs

B. Traffic Filtering Controls

All inbound and outbound network traffic should be inspected and filtered to prevent malicious communications and unauthorized services.

Proposed Tools / Technologies:

- Next-Generation Firewall (NGFW)
- Web Filtering / DNS Security
- Router Access Control Lists (ACLs)

These technologies should block:

- Peer-to-peer file sharing traffic
- Malicious or phishing websites
- Unauthorized remote desktop tools
- Suspicious ports and protocols
- Botnet command-and-control traffic



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C. Endpoint Protection

All gaming and administrative workstations should be protected against malware, ransomware, and unauthorized software installation.

Proposed Tools / Technologies:

- Business Antivirus / Endpoint Detection and Response (EDR): Bitdefender GravityZone and Kaspersky Endpoint Security
- Application Whitelisting – Microsoft AppLocker
- USB Device Control – Endpoint security suites with device control features

These tools provide:

- Real-time malware detection
- Automatic USB scanning
- Blocking of unauthorized executable files
- Centralized security monitoring

D. Data Loss Prevention (DLP)

The organization must protect billing records, customer data, and internal files from unauthorized transfer or leakage.

Proposed Tools / Technologies:

- Endpoint DLP Solutions – Microsoft Purview DLP
- USB Blocking Software – GFI EndPointSecurity
- **Cloud Backup Encryption Tools** – Acronis Cyber Protect

These controls help restrict:

- Copying sensitive files to USB devices
- Unauthorized email attachments
- Unsanctioned uploads to cloud storage

E. Intrusion Detection and Monitoring

Continuous monitoring is necessary to detect attacks, unusual behavior, or attempted compromise.



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Proposed Tools / Technologies:

- Intrusion Detection / Prevention Systems (IDS/IPS): FortiGate IPS module
- Log Monitoring / SIEM: Splunk (enterprise option)
- Network Monitoring: TP-Link Omada Controller Dashboard

These tools can detect:

- Brute-force login attempts
- Port scanning
- Malware beaconing traffic
- Device outages or suspicious bandwidth usage

F. ISP Resilience and Failover Protection

Since HG Esports Cafe utilizes multiple internet service providers, redundancy should be formally managed to maintain service continuity.

Proposed Tools / Technologies:

- Multi-WAN Router / Gateway
- Automatic Failover Monitoring – Built-in WAN health check features
- Load Balancing – Omada SDN Controller

These tools support:

- Automatic ISP failover during outages
- Balanced traffic distribution across multiple ISPs
- Improved uptime for gaming and business operations

12. Maintenance, Monitoring, and Analysis of Audit Logs

- Collect, manage, and analyze audit logs of events that could help detect, understand, or recover from an attack.

Audit log maintenance is a critical component of security monitoring for HG Esports Cafe, particularly given its hybrid infrastructure consisting of 71 endpoint devices and 2 servers. At present, the organization lacks a centralized logging mechanism, resulting in fragmented and unmonitored log data across systems. This limitation reduces the ability to detect anomalies, reconstruct incidents, and perform timely forensic investigations.



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To address this, the organization should implement centralized log collection and structured monitoring practices. Logs from endpoints, servers, and network devices should be aggregated into a single system for continuous review. Daily monitoring should be assigned to IT personnel, while weekly summaries should be submitted to management. Automated alerting for suspicious activities such as repeated failed login attempts, unauthorized configuration changes, and abnormal traffic patterns should also be enabled.

Table 15. Current Status and Proposed Procedures

Area	Current Status	Proposed Procedures
Log Collection	No centralized system across PCs and servers	Implement centralized logging for all endpoints and servers
Log Monitoring	Logs are not regularly reviewed	Assign daily monitoring and weekly reporting
Security Alerts	No automated alerting system	Enable alerts for suspicious activities and anomalies
Log Retention	No defined retention policy	Minimum 90 days local and 12 months archived storage
Log Analysis	Reactive troubleshooting only	Implement structured forensic and anomaly analysis

Table 16. Proposed Technologies and Estimated Cost

Area	Proposed Technologies	Estimated Cost
Log Collection	Windows Event Forwarding / Syslog Server	Free



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SIEM / Monitoring	Wazuh or Graylog	Free
Storage	External HDD / NAS	₱3,500–₱10,000
Dashboard	Wazuh Dashboard / Splunk Free Tier	Free

13. Controlled Access Based on the Need to Know

- Control the processes and tools used to ensure access to critical assets is granted only based on job necessity and authorization.

Controlled access within HG Esports Cafe is currently informal, with limited enforcement of structured authorization rules. Given the operational environment of 71 endpoints and shared resources, the absence of strict access control increases the risk of unauthorized access and internal misuse.

To mitigate this, a formal Role-Based Access Control (RBAC) system should be implemented. Each user should be assigned permissions strictly aligned with their responsibilities. Administrative privileges must be limited to IT personnel, while operational staff and customers should only access systems necessary for their function. Shared accounts should be eliminated to ensure accountability.

Table 17. Current Status and Proposed Procedures

Area	Current Status	Proposed Procedures
Access Model	Informal role-based access	Implement formal RBAC structure
User Accounts	Limited individual account enforcement	Require unique accounts for all users
Server Access	Broad or loosely controlled	Restrict to IT Administrator and Security Officer
Data Access	Inconsistent restrictions	Apply classification-based access rules



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Customer Access	Open workstation usage	Enforce session-based isolation per user
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Table 18. Proposed Technologies and Estimated Cost

Area	Proposed Technologies	Estimated Cost
Access Control	Windows Active Directory	Free
Permission Management	Group Policy / NTFS Permissions	Free
Identity Control	Local User Account Management	Free

14. Data Protection

- Specify the tools and methods used to protect organizational data from loss, leakage, or unauthorized access.

Data protection is essential for safeguarding operational and customer-related information within HG Esports Cafe. Currently, the organization lacks formal backup schedules, encryption standards, and centralized data governance practices. This exposes critical data to risks such as loss due to system failure, unauthorized access, and corruption.

To strengthen data protection, the organization should implement a structured backup and encryption strategy. Daily incremental backups and weekly full backups should be performed for servers and critical systems. Additionally, sensitive data should be protected using encryption tools, while USB usage should be restricted to prevent unauthorized data transfer. Cloud storage may be utilized for offsite redundancy.

Table 19. Current Status and Proposed Procedures

Area	Current Status	Proposed Procedures
Backup System	No formal backup process	Implement daily and weekly backup schedules



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Data Storage	Distributed across PCs and servers	Centralize storage on server systems
Encryption	Not consistently applied	Enable encryption for sensitive data and backups
USB Access	Unrestricted usage	Restrict and monitor USB usage
Cloud Storage	Minimal usage	Implement controlled cloud backup system

Table 20. Proposed Technologies and Estimated Cost

Area	Proposed Technologies	Estimated Cost
Backup Tools	Windows Backup / Macrium Reflect	Free
External Storage	2TB–4TB HDD	₱3,500–₱6,000
Cloud Backup	Google Drive / OneDrive	₱100–₱300/month
Encryption	BitLocker	Free

15. Incident Response and Management

- Establish structured processes for detecting, responding to, and recovering from security incidents.

Incident response management at HG Esports Cafe is currently informal and reactive in nature. There is no established Incident Response Plan (IRP), which limits the organization's ability to respond effectively to security breaches or operational disruptions.

To address this, a formal incident response framework should be implemented. This includes defining roles, establishing escalation procedures, and standardizing response actions. All incidents must follow a structured lifecycle from detection to post-incident



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review. Employees should also be trained to recognize and report security incidents promptly.

Table 21. Current Status and Proposed Procedures

Area	Current Status	Proposed Procedures
Incident Plan	Not formally documented	Develop structured Incident Response Plan
Detection	Reactive identification	Implement real-time monitoring systems
Reporting	Informal communication	Establish escalation hierarchy
Documentation	Inconsistent records	Require full incident documentation
Recovery	Ad-hoc recovery process	Standardize recovery using system backups

Table 22. Incident Response Framework

Phase	Procedures
Preparation	Training, policy development, system hardening
Identification	Detect incidents via logs and alerts
Containment	Isolate affected systems immediately
Eradication	Remove threats and vulnerabilities
Recovery	Restore systems using backups
Lessons Learned	Conduct post-incident evaluation



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Table 23. Proposed Technologies and Estimated Cost

Area	Proposed Technologies	Estimated Cost
Monitoring	Wazuh / Graylog	Free
Endpoint Security	Windows Defender / EDR tools	Free-₱200/PC/month
Backup Recovery	Macrium Reflect / Clonezilla	Free
Reporting System	Google Forms / Sheets	Free



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Appendices

Appendix A. Vulnerability Assessment Report Vulnerability Assessment Report

I. INTRODUCTION AND SCOPE

The assessment involves systematic evaluation of the security posture of participating organizations by identifying existing vulnerabilities, measuring risk exposure, and benchmarking their practices against industry peers. The assessment involves operations focusing on IT-enabled environments such as esports cafes and corporate offices. The team ensures a systematic baseline for comparing vulnerabilities, operational security, and incident management protocols across different IT environments.

The conducted assessment allows the team and focal organization to strengthen technical controls, enhance policy frameworks, and support the development of a long term security strategy.

II. ORGANIZATIONAL PROFILE

HG ESports Cafe – is a newly established business engaged in providing esports and computer rental services. It caters primarily to customers seeking access to online gaming, internet browsing, and other computer-related activities. As a developing organization, it operates in a dynamic environment where consistent service delivery and reliable connectivity are essential to meet user demands.

III. ASSESSMENT METHODOLOGY

To achieve a reliable assessment of the security posture of the four selected organizations, the group implemented a three-phase manual evaluation approach. The assessment centered on examining how each organization manages its operations,



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safeguards its physical resources, and applies technical security measures. The process was carried out through the following phases:

Survey Questionnaire: A standardized Information Assurance and Security questionnaire was administered to the representatives of each organization. Using a 4-point Likert scale, the instrument collected relevant data regarding organizational practices, security policies, and the level of implementation of security controls.

Management Interview: A follow-up interview was conducted with the representative of the main organization to support and confirm the data gathered from the questionnaire. This phase also provided additional understanding of the organization's operational activities and security-related practices.

Physical & Technical Observation: An onsite observation was performed within the primary organization to examine its working environment. This included reviewing the general setup of its systems and evaluating how physical and operational security measures are applied in practice.

IV. DATA COLLECTION PROCEDURES

Data were collected through a structured survey questionnaire and a supplementary interview with the organization's supervisor/manager. The questionnaire used a 4-point Likert scale and covered domains including IT strategy, security policies, network management, user controls, incident management, and risk management. The interview was conducted to further clarify responses and provide additional insights into the organization's operational practices and security environment.

V. KEY FINDINGS

The assessment of HG Esports Cafe reveals that while certain basic security practices are in place, several critical controls remain informal or partially implemented. Based on



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the interview and survey responses, the organization operates with updated Windows operating systems and regularly updated software, including weekly system updates. User access is also moderately controlled, as employees have role-based privileges and customers are restricted from administrative functions. However, the absence of a firewall and undisclosed antivirus solution indicates a significant gap in foundational network protection, which may expose the system to external threats.

In terms of network and user control, HG Esports Cafe demonstrates mixed implementation. The network is partially segmented, as customer devices are kept separate from the server environment, and a separate internet connection is provided for Wi-Fi users when needed. However, firewall protection is not implemented, and users are still allowed to download files under limited supervision. While a reboot-to-restore system is used to clear user data after sessions, the allowance of USB access without detailed safeguards presents a continued risk of malware introduction and system compromise.

From an operational and physical security perspective, the organization shows both strengths and limitations. The server is physically separated from client units but remains exposed within the premises, and no UPS system is in place to mitigate power interruptions. Previous incidents, such as hardware-related black screen issues, highlight vulnerabilities in system stability. Although basic controls such as router password changes and locked physical assets are implemented, the lack of documented security policies, incident protocols, and advanced protective mechanisms such as IDS/IPS or MFA suggests that the organization's overall security posture remains at a developing stage.

VI. ORGANIZATIONAL ASSETS

Table 1. Organizational Profile for HG ESports Cafe



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Assets	Hardware / Description	Quantity
Server Infrastructure	LAN Server/On-premise Servers (Locked/Restricted access)	Available
Router	TP-Link ER605 Multi-WAN Router (supports 3 ISPs, load balancing, failover)	1
Switch	TP-Link TL-SG1024D 24-Port Gigabit Switch (main distribution for PCs)	4
Network Topology	Star Topology (all PCs connected to central switch/router)	Available
Internet / ISP	PLDT-2, Converge, and Globe.	4
Hardware and Networking Components	Uninterruptible Power Supply (UPS) units	Not Available



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	Desktop PCs (Elite) (Ryzen 5 5500, 16GB RAM, GTX 1060, 27" Curved 180Hz FHD)	18
	Desktop PCs (VIP) (Ryzen 5 5600, 16GB RAM, GTX 1660 Super, 27" Curved 170Hz 2K)	25
	Desktop PCs (S-VIP) (Ryzen 5 5600, 16GB RAM, RTX 5060 OC, 27" 240Hz FHD)	27
	Monitors (Elite) (27" Curved 180Hz Full HD monitors)	18
	Monitors (VIP) (27" Curved 170Hz 2K monitors)	25
	Monitors (S-VIP) (27" 240Hz Full HD monitors)	27
	Redragon K686 HE Rapid Trigger Magnetic RGB Gaming Keyboard	70
	Headset (7.1 Badwolf Headset)	70



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	Mouse (Rapoo Gaming Mouse (Elite & VIP), Badwolf Gaming Mouse (S-VIP))	70
	Operating System (Windows 11 Home installed on all units)	70
	Alternative Telecommunication Lines	4
	Fire-prevention (Heat/Smoke detectors/Fire Alarms)	Available
	Alternative Water/Power/Cooling resources	Available
Softwares	Windows 11 home	70
Personnel	Employees	4
Security cameras/CCTV	Not disclosed.	3



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Air Conditioner	Carrier 53FLG004-ASC04H - Air Conditioner -2 CARRIER FP-53CXV030308-1 3.0hp Wall Mounted Split Type Aircon -4	6
Data Assets		

This table presents the asset profile of HG E-Sports iCafe, covering its IT infrastructure, hardware, networking components, software, and personnel. The café uses an on-premise server with restricted access, supported by a TP-Link ER605 multi-WAN router and four (4) TP-Link TL-SG1024D switches arranged in a star topology. This setup enables centralized management and load balancing across four ISPs (PLDT, Converge, and Globe) to maintain stable and reliable connectivity.

The hardware includes a tiered system of Elite, VIP, and S-VIP desktop units, along with standardized peripherals to ensure consistent performance and user experience. Environmental support systems such as air conditioning, fire-prevention mechanisms, and alternative power or cooling resources are also in place to sustain continuous operations and protect physical assets.

All units operate on Windows 11 Home and are managed by a small team of employees, providing an organized and efficient system for both technical and operational functions.

VII. RISK ASSESSMENT AND VULNERABILITY CLASSIFICATION

A. Technical & Network Vulnerabilities



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The table below presents security gaps identified within HG E-Sports iCafe's network systems and endpoint devices. These weaknesses could leave the infrastructure vulnerable to threats such as unauthorized system access, malware infiltration, and interruptions to normal service operations.

Table 2. Risk Assessment and Vulnerability Classification for Technical & Network

FINDINGS		POTENTIAL IMPACT	VULNERABILITY CLASSIFICATION	RISK CATEGORY
Black Screen Hardware Malfunction	The company experienced hardware issues.	The most damaging issue has been black screening caused by hardware malfunctions, leading to downtime and repair costs.	High	High
Multiple ISP Dependency (4 ISPs)	The shop uses two PLDT lines, one Converge, and one Globe connection.	While redundancy improves uptime, misconfiguration or lack of proper load balancing could still cause instability or downtime.	Low	Low
USB Device Access	Users are allowed to plug in their peripherals(usb devices) freely	Without strict controls, USB-borne malware could spread across machines.	High	High
Undisclosed Malicious Downloads	Users are allowed to download anything on the pc.	Customers could bypass the "ask staff first" rule by simply not informing supervisors. This opens the possibility of installing hidden monitoring tools	High	High



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		or malware that can operate during a session.		
Recent OS and Software Updates	Systems are updated.	Systems were updated recently (last week).	Low	Low
No Redundancy in Network Infrastructure	The Internet Cafe uses a single router setup.	A single router creates a critical single point of failure. If it fails, all 70 endpoints will lose network connectivity and internet access, resulting in a complete disruption of all operations.	High	High

B. Physical & Environmental Vulnerabilities

Safeguarding physical assets and maintaining proper environmental controls are essential for ensuring the reliability of the systems that support the business. The following table outlines vulnerabilities in how HG E-Sports iCafe's equipment is secured and maintained, highlighting risks that could result in hardware damage, operational downtime, or theft.

Table 3. Risk Assessment and Vulnerability Classification Physical/Environmental

FINDINGS		POTENTIAL IMPACT	VULNERABILITY CLASSIFICATION	RISK CATEGORY
Exposed Server PC	The server PC is physically exposed	though separated from client PCs. Unauthorized access or tampering could disrupt operations.	High	Critical
The shop does not	Doesn't utilize ups.	Sudden power outages can cause abrupt	High	High



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use a UPS system.		shutdowns, risking system corruption, data loss, and interruption of customer sessions.		
Locked Assets	The internet cafe secures their peripherals	Physical assets (mouse, keyboard, PC components) are locked, reducing theft risk.	Low	Low
Hardware Malfunctions (Black Screen Issue)	The company experienced hardware issues.	The most damaging issue has been black screening caused by hardware malfunctions, leading to downtime and repair costs.	High	High

C. Operational & Policy Vulnerabilities

These vulnerabilities highlight risks associated with human behavior, improper system usage, and the absence of proper procedure. These issues significantly affect the organization's ability to maintain security, accountability, and effective incident response.

Table 4. Risk Assessment and Vulnerability Classification for Operational & Policy

FINDINGS		POTENTIAL IMPACT	VULNERABILITY	RISK CATEGORY
Inadequate Server Specifications	Regular PC used as the main server instead of enterprise hardware specifications.	Using a regular PC as the main server, which creates performance bottlenecks, reliability issues and security gaps.	High	Critical



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No Redundancy in Network Infrastructure	A single router creates a critical single point of failure. If it fails, all 70 endpoints will lose network connectivity and internet access, resulting in a complete disruption of all operations.	High	High
Understaffed IT Team	Only 4 staff managing 70 workstations, server, network infrastructure, and data assets. Insufficient manpower for security monitoring, patch management, incident response, and user support limitations can create delays that increase the impact of security breaches and prolong system downtime.	High	High

VIII. BENCHMARKING ANALYSIS

The assessment of HG Esports iCafe highlights a business model centered on high-performance gaming equipment and customer experience. While the café demonstrates operational efficiency in delivering gaming services, its overall security and IT governance maturity reflects a reliance on informal practices rather than structured and standardized frameworks. This creates limitations in scalability, consistency, and long-term resilience.

A. Business Strategy



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HG Esports iCafe demonstrates the presence of business and operational practices; however, documentation remains informal and inconsistently applied. Policies exist in practice but are not fully formalized or standardized, leading to gaps in communication, enforcement, and continuity. The absence of structured documentation reduces the organization's ability to maintain consistency and adapt to future operational demands.

B. Long-Term IT Strategy

The café shows evidence of long-term IT planning, particularly in maintaining updated systems and operational continuity. However, these processes are largely informal and undocumented. While planning is consistently practiced, the lack of formal structure limits adaptability to emerging technologies and reduces the effectiveness of long-term strategic alignment.

C. Security Policies

Security measures are implemented within HG Esports iCafe, but they are not supported by formal documentation or comprehensive enforcement. Critical controls such as firewall implementation and cloud compliance are either absent or insufficient. This creates exposure to external threats and reduces the overall effectiveness of the organization's security posture.

D. Information System Security Policy

The café enforces several security practices, including maintaining updated operating systems, utilizing reboot-to-restore software, and assigning individual employee accounts with defined privileges. Despite these strengths, significant gaps remain in multi-factor authentication, USB device control, and secure cloud storage practices. These limitations increase the risk of unauthorized access and data compromise.

E. Physical Access Controls

Basic physical security measures are present, including hardware protection and general environmental controls. However, the organization lacks advanced physical access mechanisms



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such as restricted access zones and comprehensive monitoring systems. This limits the ability to fully secure critical assets and infrastructure.

F. Network Management

HG Esports iCafe demonstrates partial implementation of network security controls, including system updates and basic maintenance. However, it lacks advanced protections such as properly configured firewalls, intrusion detection systems, and secure remote access mechanisms. These gaps increase vulnerability to cyber threats and unauthorized network access.

G. Network Security Controls

HG Esports iCafe demonstrates partial implementation of network security controls, including system updates and basic maintenance. However, it lacks advanced protections such as properly configured firewalls, intrusion detection systems, and secure remote access mechanisms. These gaps increase vulnerability to cyber threats and unauthorized network access.

H. User Controls and Identity Management

User account management practices are present, particularly in password policies and employee offboarding. However, the absence of multi-factor authentication and automated account protection mechanisms weakens overall identity security. Strengthening authentication layers is necessary to reduce the risk of credential compromise.

I. Incident Management

The organization demonstrates the ability to respond to incidents and document issues as they arise. However, incident management remains reactive rather than proactive. Common issues, such as hardware malfunctions, are addressed only after occurrence, indicating a lack of preventive monitoring, testing, and formal escalation procedures.



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J. Risk Management

HG Esports iCafe recognizes the importance of risk management and conducts periodic assessments and vulnerability scans. However, execution is inconsistent and lacks formal methodology. The absence of penetration testing, structured evaluation frameworks, and integration into strategic planning leaves the organization exposed to unmitigated risks.

K. Asset Management

The café operates a physical infrastructure consisting of approximately seventy gaming stations supported by a central server. Asset management is primarily focused on maintaining performance and usability for customers. However, the lack of formal asset tracking systems, logical segmentation, and scalable infrastructure limits operational efficiency and long-term growth.

IX. Conclusion

The vulnerability assessment of HG Esports Cafe highlights a developing IT environment with several critical and high-risk security gaps, primarily driven by the absence of foundational network protections, reliance on informal policies, and limitations in infrastructure resilience. While the organization demonstrates strengths in maintaining updated systems, implementing basic user controls, and ensuring consistent service delivery through multiple ISPs, these measures are not sufficient to fully mitigate evolving cybersecurity threats. Key issues such as the lack of firewall protection, unrestricted USB access, exposed server infrastructure, and absence of a UPS system significantly increase the risk of malware infections, operational downtime, and potential data compromise.



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HG Esports Cafe shows a clear gap in terms of formalized security frameworks, advanced technical controls, and comprehensive risk management practices. The café operates effectively from a performance and customer service perspective but lacks the structured governance and layered security approach observed in more mature organizations. These findings indicate that while HG Esports Cafe is functional and growing, its current security posture remains at a developing stage and requires immediate attention.

Overall, the results of this assessment provide a critical baseline for identifying vulnerabilities and prioritizing improvements. The insights gathered will support the development of a targeted Risk Control and Action Plan, enabling HG Esports Cafe to strengthen its technical defenses, formalize security policies, and enhance long-term operational resilience against both physical and cyber threats.



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Appendix B. Interview Transcript

Exhibit A: Formal Interview Transcript

Table 5. Interview Transcript with HG Esports Cafe

Interview Questions	Client Response
Is the Windows OS and antivirus software up to date?	yes.
Does your shop use a firewall?	No.
What antivirus software are you using?	Not disclosed.
Is the main router still using the default password? (For example, was the Wi-Fi password never changed after installation?)	We already changed it.
Are users allowed to download different types of files?	yes.
Do users have any way or admin privileges that would allow them to download malware or bypass the timer?	No, only the admin can bypass it.
Over the years of operation, what has been the most frequent or most damaging issue experienced by the computer shop? (e.g., internet outages, hardware vandalism, virus infections, etc.)	I would say that the worst issue we have had so far is black screening, which was caused by hardware malfunctions.
Do all employees have access to the main server? Do they share one login, or do they have individual accounts with different privilege levels?	Not all of them, and they do not share accounts. Each account has its own privileges; for example, some accounts have hardware diagnostics access while others do not.
Do you have policies on what users are allowed or not allowed to download, tampering with PC settings, or accessing	Not disclosed.



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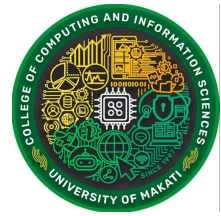
illicit content? If yes, how are these enforced by the staff?	
Is the server PC used by employees for other activities such as web browsing, watching YouTube, or downloading files?	No, it is only used for the operations of the shop.
What software is used for the PC timer?	Our own software
When was the last time the OS and software of the PCs were updated?	Last week
Do you offer Wi-Fi access to customers? If yes, is it separated from the wired network used by the server and PCs?	Sometimes, if it is really needed, for example, when they need 2FA for logging into their accounts so we provide them with temporary Wi-Fi access on their phones. We also tell them to forget the network after they are done. No, we do not allow them to connect to the same network used by the PCs; we provide a separate internet network.
Who manages the main router? (e.g., which ISP—Globe, PLDT, etc.)	We have four ISPs: two are PLDT, and the other two are Converge and Globe.
Do you use reboot-to-restore software to erase user data and open tabs after restarting the PC?	yes.
Are users allowed to plug in USB devices? If yes, how do you prevent USB-based malware from spreading?	Yes. (No further details provided)
How do you ensure that the previous user did not install hidden monitoring or screen-capture software that could steal the next user's passwords?	We first ensure that the website they are downloading from is safe. They need to ask us first before downloading such files.
Where is the server PC located? Is it secured or exposed alongside the client PCs?	It is exposed but it is separated.
How do you secure physical assets (e.g., preventing theft of mouse/keyboard or	Yes, it is locked.



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unauthorized opening of PCs to steal components)?	
Do the PCs have an Uninterruptible Power Supply (UPS)? What happens in case of a sudden power outage? How are billing and remaining user time handled?	no.

Appendix C. Photo Documentaton

Exhibit B: Photo Documentation



Figure 1. Documentation Of The Survey Questionnaire Provision for HG lcafe Comebo

Exhibit C: Specific Organizational Asset

Table 6. Specific Organizational Asset for HG E-Sports Cafe

HG E-Sports Cafe Assets List



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Assets	Hardware / Description	Quantity
Server Infrastructure	Server PC system unit (similar specs to regular PCs)	1
Network Infrastructure	On-Premise Servers	Available
Hardware and Networking Components	Router	1
	Switch	1
	Desktop PCs (Elite) (Ryzen 5 5500, 16GB RAM, GTX 1060, 27" Curved 180Hz FHD)	18
	Desktop PCs (VIP) (Ryzen 5 5600, 16GB RAM, GTX 1660 Super, 27" Curved 170Hz 2K)	25
	Monitors (S-VIP) (Ryzen 5 5600, 16GB RAM, RTX 5060 OC, 27" 240Hz FHD)	27
	Monitors (Elite) (27" Curved 180Hz Full HD monitors)	18
	Monitors (VIP) (27" Curved 170Hz 2K monitors)	25
	Monitors (S-VIP) (27" 240Hz Full HD monitors)	27
	Internet / ISP	4



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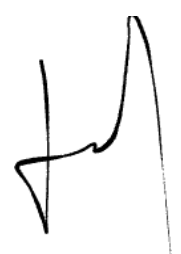
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	Keyboard (Magnetic Rapid Trigger Keyboard)	70
	Headset (7.1 Badwolf Headset)	70
	Mouse (Rapoo Gaming Mouse (Elite & VIP), Badwolf Gaming Mouse (S-VIP))	70
Software	Operating System (Windows 11 Home installed on all units)	70
Personnel	Personnel	4
Data Assets	User session data and business records	(Not Specified)

Audit Report – provide detailed accomplishment/s of each member in the group

Name	Task	Status	Signature
Robert Kenneth Valenzuela	1. Inventory of Authorized and Unauthorized Devices 2. Inventory of Authorized and Unauthorized Software 3. Secure Configurations for	Complete	



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	Hardware and Software on Mobile Devices, Laptops, Workstations, and Servers		
Raiza Nhoreen Cabalhug	1. Malware Defenses 2. Application Software Security 3. Wireless Access Control 4. Data Recovery Capability	Complete	
Mariel Francisco	1. Information Security Personnel Requirements and their Qualifications. Security Skills Assessment and Appropriate Training to Fill Gaps 2. Secure Configurations for Network Devices such as Firewalls, Routers, and Switches 3. Controlled Use of Administrative Privileges	Complete	



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	4. Boundary Defense		
Prince Andrei Briongos	1. Maintenance, Monitoring, and Analysis of Audit Logs 2. Controlled Access Based on the Need to Know 3. Data Protection. 4. Incident Response and Management.	Complete	

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